

Team Responsibilities (3 Students)

According to the guidelines, all team members must contribute visibly to the repository, experiments, and documentation. For a 3-person team, responsibilities can be divided by functional area:

Role	Primary Responsibilities	Key Deliverables
Data & Infrastructure Lead	Implement the web crawler (using tools like Scrapy or BeautifulSoup). Manage the metadata storage (e.g., PostgreSQL or Elasticsearch) and ensure compliance with robots.txt and rate limits.	Crawler code, database schema, and crawling coverage report.
NLP & ML Engineer	Develop the document classifier and ranking component. Create the labeled validation set for testing and perform precision@k and topic quality evaluations.	NLP pipeline, trained models, and quantitative evaluation results.
Systems & Ethics Architect	Design the recommendation interface and system architecture. Conduct bias, copyright, and content-quality analysis. Lead the production of the IEEE-style paper and the integrated specification.	System diagram, integrated specification table, and final research paper.

Student 1: Data & Infrastructure Lead

This student focuses on the "collection" phase of the pipeline, ensuring the system can retrieve data reliably and ethically.

- **Crawler Development:** Build the web crawler using tools like Scrapy or BeautifulSoup to navigate seed URLs and extract raw content.
- **Compliance & Ethics:** Implement logic to strictly respect `robots.txt` files and rate limits to avoid overwhelming educational sites.
- **Storage Management:** Design and maintain the database (e.g., PostgreSQL or OpenSearch) to store retrieved documents and their associated metadata.
- **Reproducibility:** Create the dependency files (Docker or Conda) and the `README.md` to ensure the environment can be recreated by others.

Student 2: NLP & ML Engineer

This student handles the "intelligence" of the project, turning raw text into categorized, ranked recommendations.

- **Classification Pipeline:** Implement Natural Language Processing (NLP) models to filter, deduplicate, and classify documents by topic or difficulty.
- **Ranking & Recommendation:** Develop the algorithm that maps educational resources to specific learner profiles or course outcomes.
- **Experimental Rigor:** Define the **baseline** (e.g., simple keyword search) and run the comparative experiments.
- **Quantitative Analysis:** Measure and report metrics such as precision@k, classification accuracy, and crawling coverage.

Student 3: Systems Architect & Ethics Lead

This student is responsible for the overall system flow, user interaction, and the high-standard academic reporting required for Tier B.

- **Architecture & UI:** Design the system architecture diagrams and build the simple recommendation interface for learners.
- **Integrated Specification:** Create the testable requirements table (8-12 requirements) and verify them through conformance checks.
- **Security & Ethical Analysis:** Lead the critical discussion on bias in recommendations, copyright implications, and learner privacy.
- **IEEE Research Paper:** Act as the lead editor for the 6-10 page research paper, ensuring all 12 required sections are included and correctly cited in IEEE style